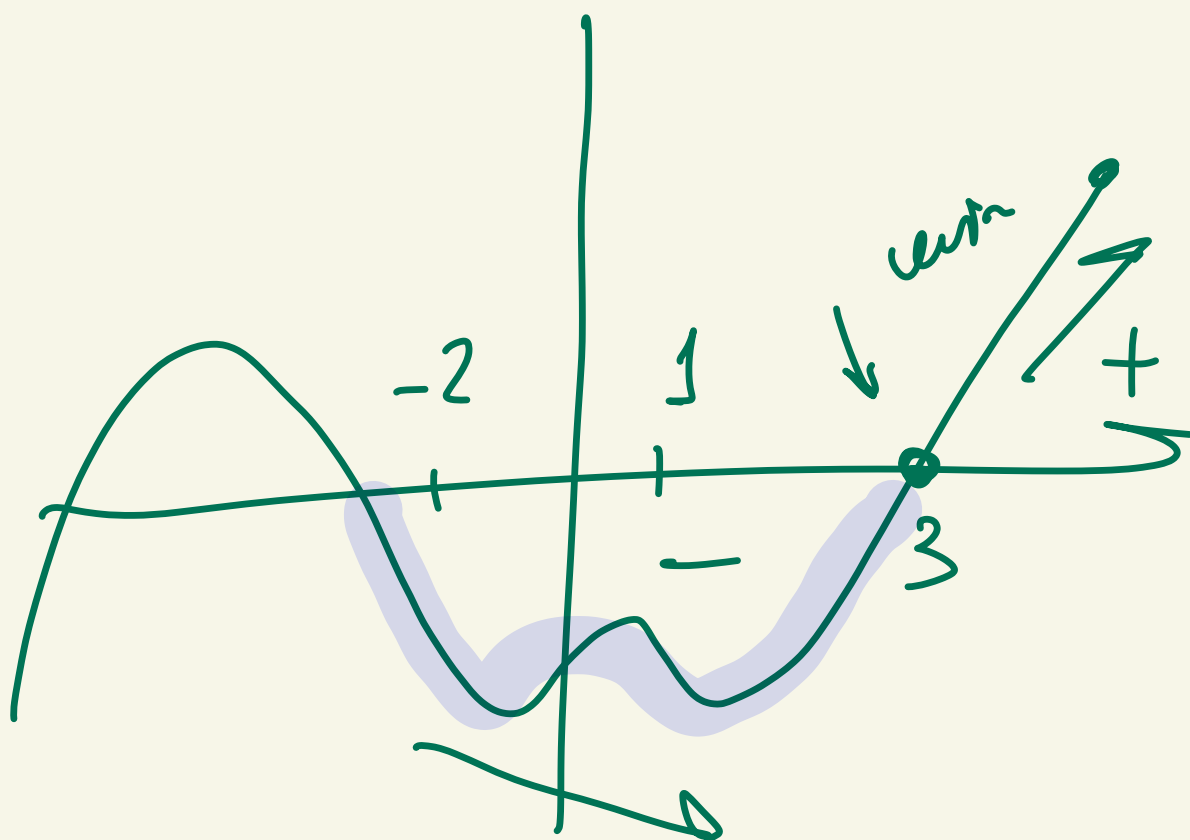
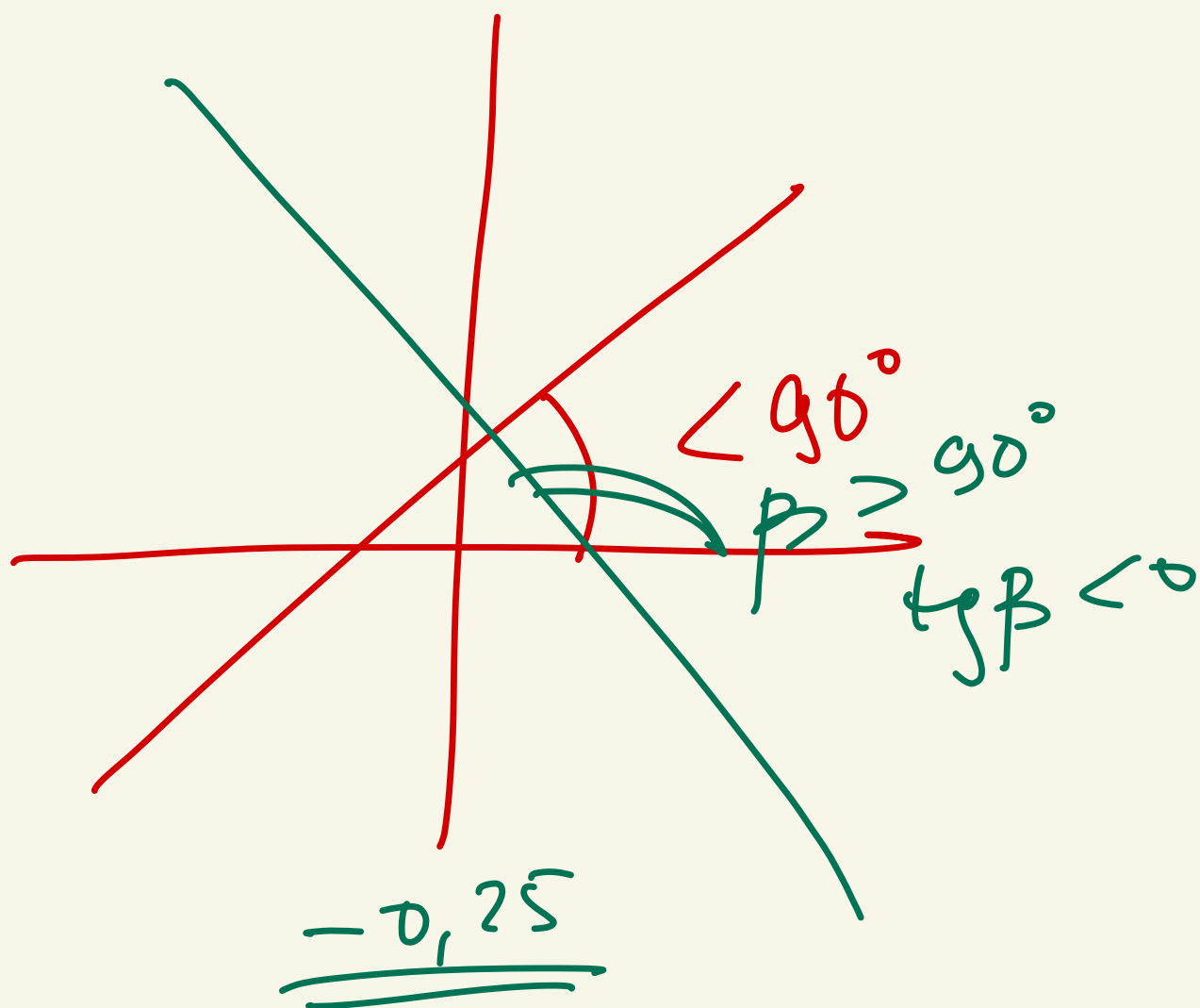








$$(-2, -2) \quad \underline{\underline{(1, 1)}}$$

$$f(x) = 2x^2 + 6x + c$$

$$f(-5)?$$

$$-1 + \log_{\sqrt{\frac{1}{2}}} 16 =$$

$$-1 + \frac{1}{\frac{1}{2}} \log_{2^{-1}} 16$$

$$-1 + 2 \cdot \log_{2^{-1}} 16 =$$

$$= -1 + 2(-1) \log_2 16 =$$

$$-1 + 2 \cdot (-1) \cdot 4 = -1 - 8 = -9$$

$$x(t) \rightarrow v(t) = \dot{x}(t) = x'(t)$$

$$a(t) = x''(t)$$

$$x(t) = 6t^2 - 48t + 17$$

$$v(t) \quad t = 9c$$

$$12t - 48 = 60$$

$$x(t) = \frac{1}{3}t^3 - 3t^2 - 5t + 3$$

$$t? \quad v = \underline{2u/c}$$

$$v(t) = t^2 - 6t - 5 = 2$$

$$t^2 - 6t - 7 = 0$$

$$\frac{6 \pm \sqrt{36 + 28}}{2} = \frac{6 \pm 8}{2} \quad \begin{matrix} 7 \\ \uparrow -1 \\ \underline{\underline{=}} \end{matrix}$$

$$h(x) = f(x) \pm g(x)$$

$$h(x) = f'(x) \pm g'(x)$$

$$\begin{matrix} x^2 + 2x \\ \downarrow \quad \downarrow \\ f(x) \quad g(x) \end{matrix}$$

$$(f(x) \cdot g(x))' = f(x) \cdot g'(x) + f'(x) \cdot g(x)$$

$$y = (x-8)e^{(x-7)}$$

min
[6; 8]

$$(x-8)e^{x-7} + e^{x-7} =$$

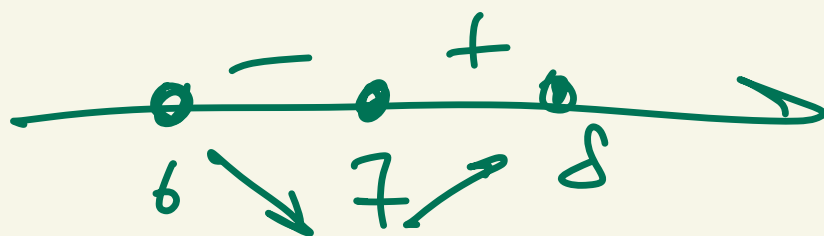
$$= e^{x-7} (x-8+1) = 0$$

x_0 $x-7=0$
 $x=7$

$$\frac{2x-2}{e}$$

$$\frac{d}{dx} (f(g(x)))' = f'(x) \cdot g'(x)$$

$$e^{2x-2} \cdot (2x-2)' = 2e^{2x-2}$$



$$y = (x+6)^2 e^{4-x}$$

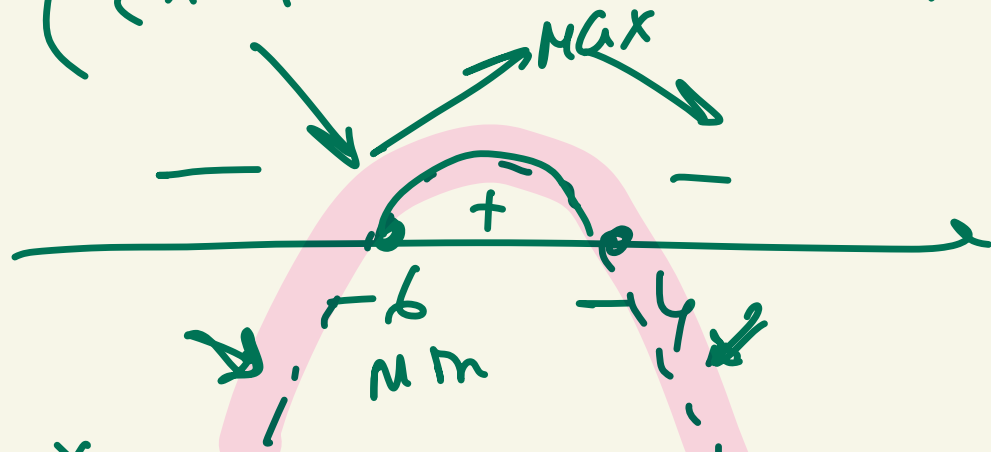
J. Max?

$$e^{4-x} (2x+12+x^2+12x+36-1)$$

$$(x^2+12x+36) e^{4-x}$$

$$(2x+12) e^{4-x} + \underline{\underline{(x^2+12x+36) e^{4-x} \cdot (-1)}}$$

$$e^{4-x} (2x+12-x^2-12x-36) = 0$$



$$e^{4-x} (-x^2 - 10x - 24)$$

$$y = (x^2 - 10x + 10) e^{5-x}$$

T. Max?

$$y = (x-2)^2 (x-4) + 5$$

$x^2 - 2 \cdot 2x + 2^2$

max
[1; 3]

$$(x^2 - 4x + 4) \cdot 1 + (x-4)(2x-4)$$

$$x^2 - 4x + 4 + 2x^2 - 4x - 8x + 16 = 0$$

$$3x^2 - 16x + 20 = 0$$

$$\frac{16 \pm \sqrt{16^2 - 240}}{2 \cdot 3} =$$

$$h(x) = \frac{f(x)}{g(x)}$$

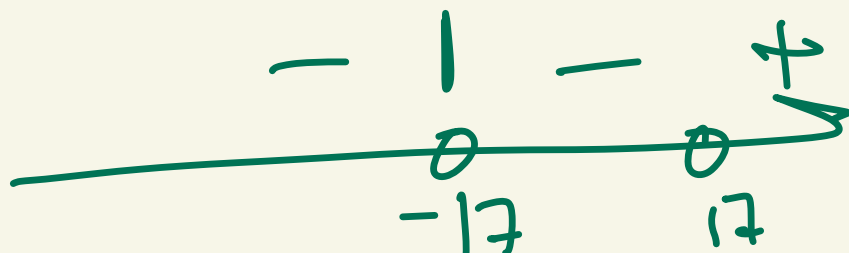
$$h'(x) = \frac{f'(x) \cdot g(x) - g'(x) f(x)}{g^2(x)}$$

$$y = \frac{x^2 + 25}{x}$$

Намб. 31
[-10; -1]

$$y = \frac{-x}{x^2 + 289}$$

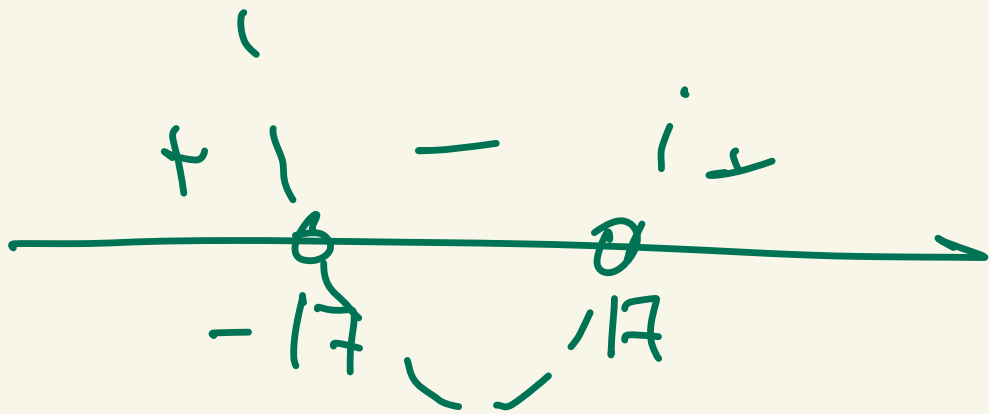
T. max?



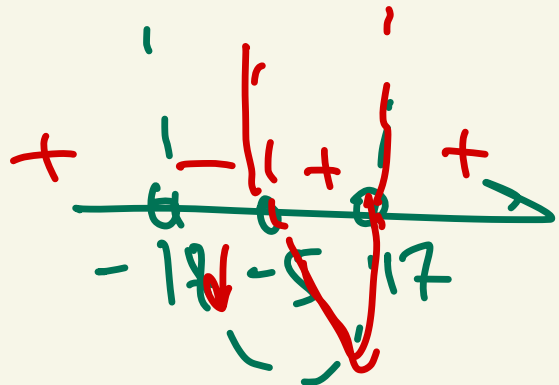
$$y' = \frac{-x^2 - 289 + 2x^2}{(x-17)(x+17)}$$

$$y' = \frac{-1(x^2 + 289) + 2x(-x)}{(x^2 + 289)^2} =$$

$$\frac{x^2 - 289}{(x^2 + 289)}$$



$$\frac{x^2 - 289}{(x-17)(x+5)}$$



$$y = (f(g(x)))' = f'(g(x)) \cdot g'(x)$$

$$y = \ln(\sqrt{x^2 + 2x})$$

$$(\ln x)' = \frac{1}{x}$$

$$(\ln t)' = \frac{1}{t} t'$$

$$x^{\frac{1}{2}} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$y' = \frac{1}{\sqrt{x^2 + 2x}} (\sqrt{x^2 + 2x})' =$$

$$\frac{1}{\sqrt{x^2 + 2x}} \cdot \frac{(x^2 + 2x)'}{2\sqrt{x^2 + 2x}} = \frac{2(x+1)}{2(x^2 + 2x)}$$

$$= \frac{x+1}{x(x+2)}$$

$$y = \ln(x^2 + 4x + 4)$$

$$y' = \frac{1}{x^2 + 4x + 4} \cdot (2x + 4)$$

$$y = 3x - \ln(x+3)^3$$

Наиб. зт

$$3 - \frac{1}{(x+3)^2} \cdot 3(x+3)^2 \quad [-2,5; 0]$$

$$3(x+3)^2 \cdot (1)$$

$$y' = 3 - \frac{3}{x+3} = 0 \rightarrow x = -2$$

$y \neq (x)$

$y(-2)$

Найдите наименьшее значение функции $y = 3x - \ln(x+3)^3$ на отрезке $[-2,5; 0]$.

$$y = -6 - \ln(-2+3)^3 = -6 - \ln 1 = -6$$

$$\ln(x+5)^5 - 5x = y$$

max. 3M

$[-4,5; 0]$

$$y' = \frac{1}{(x+5)^5} \cdot 5(x+5)^4 - 5$$

$$\frac{5}{x+5} - 5 =$$

$$x = -0,8$$

$$\frac{5 - 5(x+5)}{x+5} = 0$$

$$\frac{5 - 5x - 25}{x+5} = 0$$

$$-5x$$

$$y = e^{2x} - 6e^x + 3$$

Начин. 3 и

$\phi - \text{un}$

$[1; 3]$

$$y' = e^{2x} \cdot 2 - 6e^x =$$

$$e^x \cdot e^x \cdot 2 - 6e^x =$$

$$0 \neq e^x (e^x \cdot 2 - 6) = 0$$

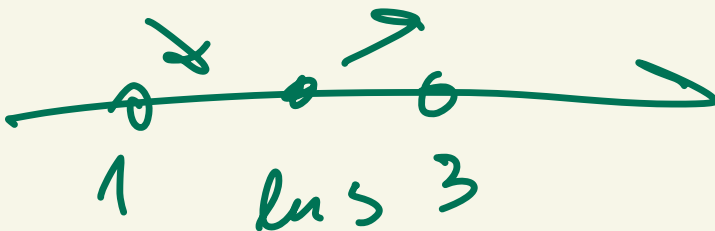
$$2e^x = 6$$

$$e^x = 3$$

$$e \approx 2,71$$

$$\ln e^x = \ln 3$$

$$\underline{\underline{x = \ln 3}}$$



$$5^x = 10 \rightarrow \log_5 5^x = \log_5 10$$

$$e^{2 \ln 3} - 6 e^{\ln 3} + 3 = 0$$

$$\log_b a = c$$

$$e^{\ln 9} - 6 \cdot 3 + 3$$

$$9 - 18 + 3 = -6$$

$$y = e^{4x} - 4e^x + 8$$

Heim
3H

$$y(0) = 1 - 4 + 8 = 5 \quad [-2; 2]$$

$$y' = 4e^{4x} - 4e^x =$$

$$e^{4x} = e^x \cdot e^{3x}$$

$$\boxed{\log_e 1 = 0}$$

$$4e^x \cdot e^{3x} - 4e^x =$$

$$= e^x (4e^{3x} - 4)$$

~~0~~

$$4e^{3x} - 4 = 0$$

$$e^{3x} = 1$$

$$\ln e^{3x} = \ln 1 = 0$$

$$3x = 0$$

$$x = 0$$

$$y = e^{2x} - 8e^x + 9$$

Минимум ?

$$2e^{2x} - 8e^x$$
$$= e^x(2e^x - 8)$$

$$2e^x = 8$$

$$e^x = 4$$

$$x = \ln 4$$

$$e^{2\ln 4} - 8e^{\ln 4} + 9$$

$$4^{2\ln 2} - 8 \cdot 4 + 9$$

$$16 - 32 + 9 = -7$$